

Endomorphisms

Gian Carlo

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1 Introduction

Treating the algebraic structure of an endomorphism set on some finite set S as a monoid seems rather boring to me. As with the portuguese texts, this will make mixed use of the language persons.

2 Investigation

Let S be a finite set. The endomorphism set on S is the set of all functions $S \rightarrow S$. I will denote this endomorphism set by E_S .

Among the elements of E_S are the bijective functions. Those form a known group named the permutation group of S . I will denote this group by P_S . Let $n = |S|$. There are $n!$ such bijections. That leaves us $n^n - n!$ elements to structurize.

Another set of elements of E_S are functions of the form $f(x) = s_0$ for a given s_0 in S . Since there are n options for such s_0 , there are n functions of this type. Let us define an operation $\cdot$ on E_S given by

$$f \cdot g = g \circ f. \quad (1)$$

This is so we can read products left-to-right like usual. For the elements of the form $f(x) = s_0$, lets call them R_S . For every $r \in R_S$ and any product $P = f_1 f_2 \dots f_p$ of $f_i \in E_S$ we have

$$Pr = r. \quad (2)$$

That is because $r(x) = s_0$ for whatever input and as such it does not matter what is done to the input beforehand, so long as it remains in S . In particular, every $r \in R_S$, when placed at the end of a product, behaves like 0 does for usual multiplication. Since the end of the product is at the rightmost symbol, we name these elements the rightmost zeroes, or just R_S .

We will denote the identity function by $e(x) = x$ or just e . So $eP = Pe = e$ for any product P as before. Since each $r \in R_S$ has only one image, we will instead denote it by $r_0 \in S$.

Take $r \in R_S$. If we consider the equation

$$rx = e \quad (3)$$

we could apply some input s to both sides and get

$$x(r(s)) = e(s) \quad (4)$$

and since $r(s) = r_0$, we get $x(r_0) = e(s) = s$. Thus $x(r_0)$ has to be any element of S . This is a relation on S . We will look at a specific type of relation.

Let $t \subseteq S \times C$ be a relation on S . We will call t a **branching** iff

1. For all $c \in C$ there is $s \in S$ such that $(s, c) \in t$; And
2. if $(s, c) \in t$ and $(z, c) \in t$ then $s = z$.

We shall denote $(s, c) \in t$ by $c(t) = s$. This looks well defined, though I will not rigorously check. With branchings, we can complete R_S . In particular, the branching given by $r_0(t) = s, \forall s \in S$ tells us that $rt = e$. So now every $r \in R_S$ has an inverse branching, if we assume composition is done as expected.

It seems we can even complete E_S as a group with branchings. Let us see. We need no inverses for e and elements of P_S . So let $f \in E_S$ be a function not equal to these. Have any $s \in S$ and consider

$$f^{-1}(s) = \{x \in S : f(x) = s\}. \quad (5)$$

Clearly the branching given by

$$s(t_f) = x, \forall x \in f^{-1}(s) \quad (6)$$

gives us an inverse to f . For confirmation, if $f(x) = y$, then $y(t_f) = x$ and as such $ft(x) = (f(x))(t_f) = y(t_f) = x = e(x)$.

Another thing we can do is consider near-inverses. Say we take $f \in E_S$ such that $f(x) = s_0 \in S$. We can not invert this with another function. But we can invert it for every $x \in S$. Let $z \in S$. The inverse of $f(z) = s_0$ is given by $g(s_0) = z$. We can always find such a function $g \in S$. What we lose is uniqueness of inverse. For simplicity, let's restrict this even more by forcing g to behave as e everywhere else but s_0 .

2.1 A step back

It seems more prudent trying to understand the endomorphism set in its usual form rather than trying some form of group completion. I like the name endomorphism but I will allow myself to also use the name "function set" to describe it.

So we know constant functions of the form $f(x) = a$ are right-zeros in the product, in the sense they determine the whole of the product as themselves by simply being a factor in it at the very end.

Well, there are as many right-zeroes as there are elements of the set, because for each $c \in S$ we can separate the function $f(x) = s$ and count it as one. The total number of functions that exist in a set S to itself is known to be, for $n = |S|$, n^n . It is also known that there are $n!$ bijections and n right-zeroes. So we got a nice little sum to break down

$$n^n = n! + n + R$$

where R is hiding the other amounts of types of function.

We could elaborate on all the summands hidden in R through combinatorics. For that, let us say that a function is an k -classification if its image has cardinality k (I think of it as the function classifying the set into k classes). Let us restrict ourselves to finite sets for now. We know that there are $n!$ n -classifications and n 1-classifications.

3 Synthesis

Definition 3.1 (Branching): Let A and B be two non empty sets. A relation $R \subseteq A \times B$ obeying:

1. For all $b \in B$ there is an $a \in A$ such that $(a, b) \in R$; and
2. For all $a, e \in A$ and $b \in B$, if $(a, b) \in R$ and $(e, b) \in R$ then $a = e$;

will be called a **branching** of A to B . ○

Note 3.2: We will write $(a, b) \in R$ as $b|R = a$. This can be read as b comes from a by R . The second condition for branching tells us this notation is well defined.

We will also write a branching $R \subseteq A \times B$ as $R : B \leftarrow A$. One way to read this is: by R , what is in B comes from A . A better way is A branches from B , which I like more because it follows the arrow. △

Definition 3.3 (Composition of Branchings): Let A , B and C be three non empty sets. If there are branchings R, S such that $R : B \leftarrow A$ and $S : C \leftarrow B$, then the **composition** of R and S , denoted by RS , is given by the equation

$$x|RS = (x|S)|R \tag{7}$$

for all $x \in C$.

○

Definition 3.4: Let E_S be the set of functions of $S \rightarrow S$, and let $f \in E_S$ be a constant function $f(x) = s_0 \in S$ for all $x \in S$. A function r is a **right near inverse** of f if there exists $z \in S$ such that

$$\begin{cases} r(x) = s_0, & x = z \\ r(x) = x, & x \neq z \end{cases} \quad (8)$$

for all $x \in S$.

○

Proposition 3.5: Let S be a non-empty set of known cardinality C , and E_S the set of function of $S \rightarrow S$. If $f \in E_S$ then there exists exactly C right near inverses of f .

Proof: TBD.

■

4 Conclusions

How does one conclude something that is ongoing if not by ending it abrupte